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import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression

from sklearn.model_selection import train_test_split


data = pd.DataFrame({
    'X': [1, 2, 3, 4, 5],
    'Y': [2, 4, 5, 4, 5]
})


X = data[['X']]
Y = data['Y']


X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=0)


model = LinearRegression()


model.fit(X_train, Y_train)


Y_pred = model.predict(X_test)


plt.scatter(X, Y, color='blue')


plt.plot(X, model.predict(X), color='red')


plt.title("Linear Regression")
```

```
plt.xlabel("X")
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```
plt.ylabel("Y")
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```
plt.show()
```

